



DEDAN KIMATHI UNIVERSITY
OF TECHNOLOGY
UNIVERSITY EXAMINATIONS 2025/2026
THIRD YEAR SECOND SEMESTER EXAMINATION FOR BACHELOR OF SCIENCE
DEGREE IN MECHATRONIC ENGINEERING
EMT 3205 MEASUREMENT AND INSTRUMENTATION

DATE: 11th FEBRUARY 2026

11.00AM-1.00PM

TIME: 2 HOURS

INSTRUCTIONS

1. This paper consists of FIVE questions.
2. You are required to answer THREE questions only.
3. Question **ONE** is compulsory.
4. Attempt ONE question from section **A** and another ONE question in section **B**.
5. Question ONE carries 30 marks and the others carry 20 marks each.

Question One (30 marks)

- a) Name and briefly explain the two main types of errors that arise during the measurement process. [4 marks]
- b) State four static characteristics of measurement instruments commonly encountered in engineering applications. [4 marks]
- c) Name four types of AC bridges and state the primary measurement quantity each is used to measure. [4 marks]
- d) Distinguish between zero drift and sensitivity drift, explaining how each type of disturbance affects the measurement output differently. [6 marks]
- e) List three examples of resistive transducers and state what physical quantity each measure. [3 marks]
- f) Explain the difference between repeatability and reproducibility [2 marks]
- g) Name three characteristics that distinguish digital instruments from analog instruments. [3 marks]
- h) List the two main types of instrument transformers and state the primary purpose of each. [4 marks]

SECTION A (Answer only ONE question)

Question Two (20 marks)

a) A spring balance is being evaluated for use in a pharmaceutical production facility. The device was calibrated at 22°C with the following performance:

- Load of 50 kg → Deflection of 20 mm
- Load of 100 kg → Deflection of 40 mm
- Load of 150 kg → Deflection of 60 mm

During winter operation, when the facility temperature drops to 5°C, repeat measurements of a 100 kg standard weight produce the following deflections:

- Measurement 1: 39.2 mm
- Measurement 2: 39.1 mm
- Measurement 3: 39.3 mm
- Measurement 4: 39.0 mm

i. Calculate the calibration sensitivity of the spring balance at 22°C.

[2 marks]

ii. Determine whether zero drift, sensitivity drift, or both are present at 5°C. Show your calculations.

[3 marks]

iii. For pharmaceutical production where accuracy is critical, discuss what measurement error types have occurred and recommend what corrective action should be taken.

[3 marks]

b) Name and briefly explain four types of remote sensing technologies and provide one application example for each.

[12 marks]

Question Three (20 marks)

a) State and explain **five main applications** of microprocessors in instrumentation systems.

[10 marks]

b) A mechatronic engineer is designing a vibration monitoring system for a robotic arm in a manufacturing cell. The vibration sensor outputs a weak analog signal of 8 mV, but the data acquisition (DAQ) system requires an input between 0.8V and 2.0V. The sensor signal also contains high-frequency electrical noise from nearby welding machines, and the robot operates in an electrically noisy environment where voltage spikes may occur.

i. Determine the amplifier gain needed to raise the 8-mV signal to 0.8 V minimum.

[2 marks]

ii. Select a suitable filter type and cutoff frequency to remove the high-frequency noise if the useful vibration frequencies are 0–200 Hz.

[2 marks]

iii. Calculate the minimum isolation resistance required to limit current to 0.5 mA during a possible 300 V spike.

[3 marks]

- iv. Explain why linearization is NOT required for this sensor and justify your answer. [3 marks]

SECTION B (Answer only ONE question)

Question Four (20 marks)

- a) Explain why systematic errors are considered more problematic than random errors in measurement systems, especially in applications such as quality control. [4 marks]
- b) Define calibration and explain its four benefits in instrumentation. [10 marks]
- c) A sensor is calibrated to measure temperature in the range of 0°C to 100°C . The sensor's output is assumed to follow a linear relationship between input and output, but the actual output deviates from the ideal linear curve at certain points. The data is as follows:

Input Temperature $^{\circ}\text{C}$	Ideal Output (mV)	Actual Output (mV)
0	0	0
25	250	245
50	500	490
75	750	765
100	1000	1020

Determine the **maximum non-linearity error** as a percentage of the full-scale output. [6 marks]

Question Five (20 marks)

- a) An industrial plant uses several analog instruments to measure electrical and mechanical parameters. The engineer needs to select the appropriate instruments for various tasks based on the following requirements.
- i. A **direct measurement** of the electrical current is needed for monitoring the plant's power supply. The instrument must provide a reading on a calibrated scale by the deflection of a pointer. **Which type of instrument is best suited for this application, and why?** [3 marks]
- ii. A **precision measurement** of the electrical resistance is required to detect slight variations in material conductivity. The instrument should provide accurate results by balancing the unknown resistance with a known standard, showing zero when the values are equal. **Which type of instrument would be most suitable for this application, and why?** [3 marks]
- iii. The plant engineer needs a **general-purpose instrument** to measure the temperature of a machine part. The instrument should be simple and inexpensive. [3 marks]

providing a reliable reading of temperature without needing extreme precision. Which instrument type would you recommend for this task, and why?

[3 marks]

- iv. A **comparison-based measurement** is required to measure an unknown voltage by comparing it with a known standard voltage. The instrument should display the difference between the unknown and standard voltage. Which type of instrument would be ideal for this application, and why?

[3 marks]

- b) A thermistor is used to measure temperature, but its output voltage V is non-linear with respect to temperature T . The relationship between voltage and temperature for the thermistor can be approximated by the equation: $V = 0.01T^2$ where V is in volts and T is in degrees Celsius. To make this signal usable for a linear system, you need to apply a linearization method that adjusts the output to be directly proportional to temperature.

- i. Calculate the adjusted output V_{linear} that would provide a linear relationship between V_{linear} and T .

[2 mark]

- ii. Determine V_{linear} at $T = 10^\circ\text{C}$ and $T = 20^\circ\text{C}$.

[2 mark]

- c) Given a measurement system exposed to electromagnetic interference, how would you apply shielding and grounding techniques to reduce noise?

[4 marks]